Rules for data

Before any of our data goes near a model: what may go where, and how we make sure it doesn’t go anywhere else.

60 min | Recorded | Class 8 of 15

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Section 1 of 6

Why this comes first

Everything from here on sends data somewhere.

01

Three companies, three very different attitudes

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Every prompt **leaves the building**, unless it doesn’t.

A prompt is a copy of our data, sent to someone else’s computer, unless we decide otherwise.

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Three companies, three very different attitudes

We’re in the middle, and we work with both ends.

01 LOL
Fairly relaxed. If a tool is in the directory, use it up to its limit.

02 B.U.M.
No written AI policy. Microsoft 365 Copilot is switched on in Office, and that’s the extent of it.

03 A.S.S.
Strict. Background check data doesn’t leave approved systems, ever.

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Section 2 of 6

Our policy at LOL

A directory of tools, each with a ceiling.

01

We sort data into five classes, from Public to Restricted

02

Every approved tool has a ceiling; nothing above it goes in

03

Four tickets from this month, and where each one may go

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We sort data into five classes, from Public to Restricted

1. **LOL Public**
2. **LOL Internal**
3. **LOL Confidential**
4. **LOL Highly Confidential**
5. **LOL Restricted**

Payslips, bank details and background checks sit at the top. Shipment tracking for a customer who already knows about it sits near the bottom.

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Every approved tool has a ceiling; nothing above it goes in

Lighthouse, our internal AI gateway	up to LOL Restricted
ServiceDesk AI, built into the ticketing tool	up to LOL Highly Confidential
Open models running on your own laptop	up to LOL Confidential
Anything else	LOL Public only

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Four tickets from this month, and where each one may go

TICKET	WHAT IT'S ABOUT	CLASS	ALLOWED TOOLS
TKT-418141	Armadillo returned two driver checks as "inconclusive"	ASS Restricted	No AI tool at all
TKT-418212	A deduction on one employee's payslip	LOL Highly Confidential	ServiceDesk AI or Lighthouse
TKT-418087	Customs docs rejected for a Kestrel shipment	LOL Internal	Any directory tool
TKT-418255	Suspicious bank-change email from "Drymark"	LOL Confidential	Lighthouse, or a local model

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Section 3 of 6

The people we work with

Their data comes with their rules.

01

Bayerische Unfug Maschinen: no written policy

02

Armadillo Security Services: strict, and their rules follow their data into our systems

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Bayerische Unfug Maschinen: no written policy

No written AI policy. Microsoft 365 Copilot is switched on in Office, and that’s the extent of it.

01 What that means for them

Their staff can put our rate cards or our emails into Copilot, and nothing stops it.

02 What that means for us

Nothing changes on our side. Treat anything BUM sends us as LOL Confidential, whatever they did with it before.

03 What we don’t do

Assume their shipping documents are fine to paste anywhere because they’re “only BUM’s”.

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Armadillo Security Services: strict, and their rules follow their data into our systems

ASS Public
ASS Internal
ASS Confidential
ASS Restricted

- Approved AI tools take ASS Confidential at most.
- Unapproved AI tools are blocked.
- If one gets opened anyway, it runs in browser isolation: nothing copies in or out.

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Section 4 of 6

For reference: the real ones

The policies that apply to the people in this room.

01

The real policies look a lot like the ones we made up

02

Where the public frameworks sit

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The real policies look a lot like the ones we made up

TCS

- Four classes: TCS Public, Internal Use, Confidential, Restricted.
- Approved AI tools: TCS Confidential at most.
- Unapproved AI tools are banned. If opened, they run in Zscaler Browser Isolation, with no data movement.

Cisco

- A directory lists every approved tool and the highest class it may receive.
- CircuIT: Cisco Restricted. Jira AI services: Cisco Highly Confidential.
- Local models on a laptop: Cisco Confidential. Everything else: Cisco Public.

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Where the public frameworks sit

EU AI Act	Sorts AI uses by risk, from banned to minimal. Obligations are being phased in from 2025.
NIST AI RMF	A voluntary US framework: govern, map, measure, manage. Good as a checklist.
ISO/IEC 42001	A certifiable management system for AI, like ISO 27001 for security.
India’s DPDP Act	Personal data protection. Applies to employee and customer data we hold.

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Section 5 of 6

Making it stick

Redact before the prompt, and a gate that says no.

01

Every call to a model goes through the same four checks

02

The same rules apply to training data, not only to prompts

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Every call to a model goes through the same four checks

1

Classify the data, or refuse

2

Redact what the tool may not see

3

Check the tool’s ceiling

4

Log what was sent, and where

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The same rules apply to training data, not only to prompts

- Ticket history has payslip complaints, bank details and background checks in it
- A model trained on it can repeat what it saw
- Redact before training, the same way as before prompting

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Section 6 of 6

Code walkthrough

Redaction and the gate.

01

Redaction swaps what must not leave for a placeholder the model can still reason about

02

The gate refuses anything unclassified, and anything above the tool’s ceiling

03

Regex redaction catches the shapes, and misses everything else

04

This class isn’t a place on the card; it’s the gate in front of places 3 to 6

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Redaction swaps what must not leave for a placeholder the model can still reason about

code/class08/redact.py

```
RULES = [
    (re.compile(r"[\w.+-]+@[\w-]+(?:\.[\w-]+)+"), "<email>"),
    (re.compile(r"\b[A-Z]{2}\d{2}(?:\s?[A-Z0-9]{1,4}){3,8}\b"), "<iban>"),
    (re.compile(r"\bE\d{6}\b"), "<employee-id>"),
    (re.compile(r"\bCW-\d{5}\b"), "<contractor-id>"),
    (re.compile(r"\+?\d[\d\s-]{8,}\d"), "<phone>"),
]
```


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The gate refuses anything unclassified, and anything above the tool's ceiling

code/class08/gate.py

```python
def allowed(tool: str, data_class: str | None) -> bool:
    if data_class not in LEVELS:                    # unclassified: refuse, don't guess
        return False
    limit = DIRECTORY.get(tool, "LOL Public")       # anything not in the directory
    return LEVELS.index(data_class) <= LEVELS.index(limit)
```


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Regex redaction catches the shapes, and misses everything else

It finds emails, IBANs, employee and contractor IDs, phone numbers. It won’t find “the driver whose check came back inconclusive” or a name typed in lowercase.

So redaction lowers the risk. The gate and the data class decide whether the text goes at all.

Before

Employee E204518, +44 7700 900123, DE89 3704 0044 0532 0130 00

After

Employee <employee-id>, <phone>, <iban>

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This class isn’t a place on the card; it’s the gate in front of places 3 to 6

1. A script	The steps are known and don’t change.
2. A rule	The decision is known and must be explainable line by line.
3. Classical ML	Labelled history, and the input is a table or short text.
4. A small trained model	Lots of text, images or audio, enough labels, cost per call matters.
5. An LLM with a prompt	Varied language in and out, few labels, someone checks the output.
6. An LLM plus retrieval, tools or agents	Needs private knowledge, has to act, or takes several steps.
7. Not yet	No ground truth, needs a guarantee, or nobody owns the result.

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What to do next

1. Find three tickets in the handwritten set that must never reach an AI tool, and say why.
2. Add a pattern to `redact.py` for shipment IDs, then decide whether it should be there.
3. Read the class 9 pre-reading on embeddings.

Questions after class: the course channel.

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